FoodRescue TN — Microservices architecture

Angular frontend • Spring Boot 3 + Spring Cloud • REST/OpenFeign + RabbitMQ • Docker Compose

PROPOSED SYSTEM DESIGN

PUBLIC NETWORK

Users and Angular UI

ANGULAR

Responsive web app

HTTPS

USER ROLES

- Donors & associations
- Volunteer drivers
- Citizens & admin

Sign in via Keycloak

Bearer JWT after login

HTTPS to API Gateway

SPRING CLOUD API GATEWAY

REST routing

JWT validation

Service discovery

/api/* entry point

Gateway

Spring Boot

PRIVATE NETWORK

Backend and infrastructure

DOCKER COMPOSE / MICROSERVICE BACKEND

API Gateway routes HTTP requests to ALL seven services (e.g. /api/offers, /api/matches, /api/drivers...)

MS1 · Donation Offer

Spring Boot 3

Surplus food, stock reservations

MongoDB · isolated data

MongoDB

Reserve stock

MS2 · Matching & Pickup

Spring Boot 3

Charity matches, pickup assignments

PostgreSQL + PostGIS

PostgreSQL

PostGIS

Find charity

MS3 · Partner Registry & Trust

Spring Boot 3

Verified donors and charities

PostgreSQL · isolated data

PostgreSQL

Reserve

Available drivers

MS6 · Surprise Basket

Spring Boot 3

Discounted baskets and orders

MongoDB · isolated data

MongoDB

MS5 · Volunteer Driver

Spring Boot 3

Vehicles, availability, ratings

PostgreSQL · isolated data

PostgreSQL

MS4 · Impact Analytics

Spring Boot 3

Rescued kg, forecasts, CO_2e

PostgreSQL + Redis cache

PostgreSQL

Redis cache

Impact updates

Offer event

BasketCollected

Match / delivery events

RabbitMQ / asynchronous event bus

OfferPublished · MatchConfirmed · DeliveryCompleted · BasketCollected

Notifications and impact updates are processed without blocking the request.

Notify

COMMON · User & Notifications

Spring Boot 3 • PostgreSQL

Profiles, preferences, notifications

SHARED INFRASTRUCTURE

EUREKA

Service discovery

CONFIG SERVER

Centralised configuration

KEYCLOAK

Identity, OAuth2 / JWT

DOCKER COMPOSE

Local containers and networks

Each microservice owns its database • Eureka + Config Server support service startup and routing • Keycloak issues JWTs

LEGEND

HTTP through gateway

REST / OpenFeign (sync)

RabbitMQ (async)

FoodRescue TN • ESPRIT AWD • Architecture proposal • Angular confirmed